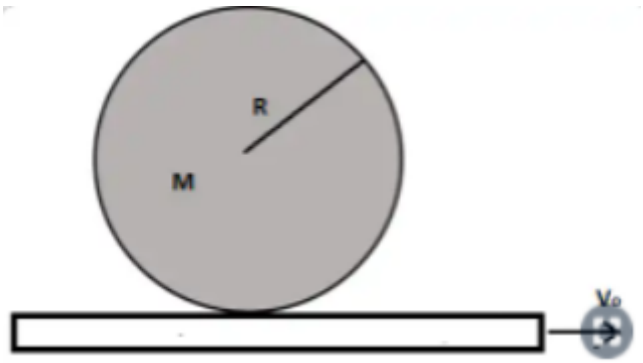
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A solid sphere of mass m and radius R is gently placed on a conveyer belt moving with constant velocity v_0. If coefficient of friction between belt and sphere is $2/7$ the distance traveled by the centre of the sphere before it starts pure rolling is



(A) $\frac{\mathrm{v}_0^2}{7\,\mathrm{g}}$

(B) $\frac{2v_0^2}{49g}$

(C) $\frac{2\mathrm{v}_0^2}{5\,\mathrm{g}}$

(D) $\frac{2\mathrm{v}_0^2}{7\,\mathrm{g}}$

Q.2

Find the moment of inertia of a thin uniform rod of mass M and length L about an axis passing through its centre and making an angle θ with the rod.

(A) $\frac{ML^2\cos\theta}{12}$

(B) $\frac{ML^2\sin^2\theta}{12}$

(C) $\frac{ML^2\cos\theta}{18}$

(D) $\frac{ML^2\sin^2\theta}{18}$

Q.3

The radius of gyration K of a hollow sphere of mass M and radius R about a certain axis is equal to R. Find the distance of that axis from the centre of the sphere.

(A) $R/\sqrt{2}$

(B) $R/\sqrt{3}$

(C) $R/\sqrt{7}$

(D) $R/\sqrt{5}$

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Q.4

A wheel having angular momentum $2\pi \text{kg} - \text{m}^2/\text{s}$ about its vertical axis, rotates at the rate of 60rpm about this axis, The torque which can stop the wheel's rotation in 30 sec would be

(A) $\frac{2\pi}{15}\text{Nm}$

(B) $\frac{\pi}{18}\text{Nm}$

(C) $\frac{\pi}{12}\text{Nm}$

(D) $\frac{\pi}{15}\text{Nm}$

Q.5

A wheel having moment of inertia $2\ \text{kg} - \text{m}^2$ about its vertical axis, rotates at the rate of 60rpm about this axis, The torque which can stop the wheel's rotation in one minute would be

(A) 1.12Nm

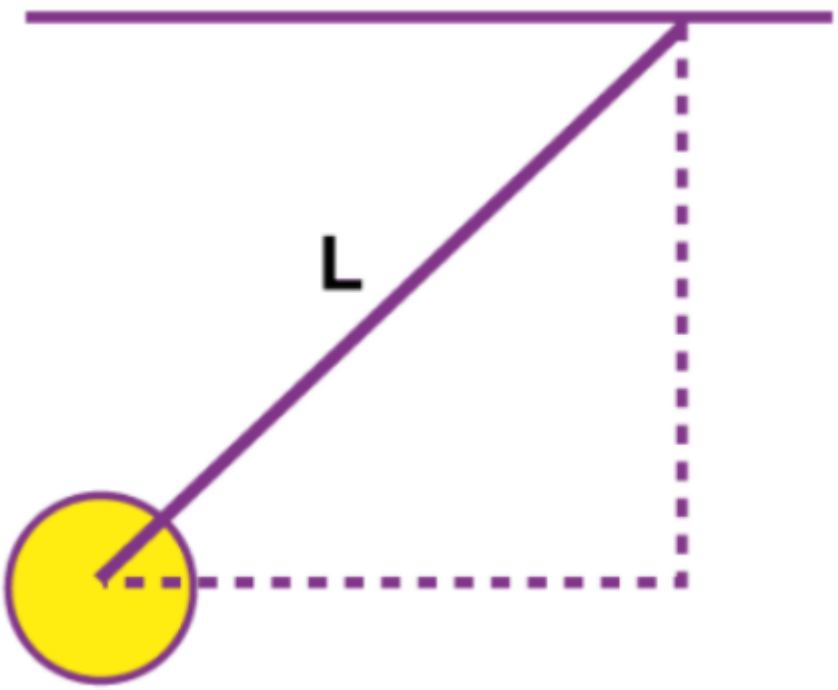
(B) 0.83Nm

(C) 0.55Nm

(D) 0.21Nm

Q.6

A ball of mass $(\text{m})0.5\ \text{kg}$ is attached to the end of a string having a length $(\text{L})0.5\ \text{m}$. The ball is rotated on a horizontal circular path about a vertical axis. The maximum tension that the string can bear is $324\ \text{N}$. The maximum possible value of the angular velocity of ball (in radian/s) is



(A) 9

(B) 18

(C) 27

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(D) 36

Q.7

A couple produces

(A) purely linear motion

(B) purely rotational motion

(C) linear and rotational motion

(D) no motion

Q.8

A wooden cube is placed on a rough horizontal table, a force is applied to the cube. Gradually the force is increased. Whether the cube slides before toppling or topples before sliding is independent of :

(A) the position of point of application of the force

(B) the length of the edge of the cube

(C) mass of the cube

(D) Coefficient of friction between the cube and the table

Q.9

A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K. If radius of the ball be R, then the fraction of total energy associated with its rotational energy will be

(A) $\frac{\mathrm{K}^2}{\mathrm{R}^2}$

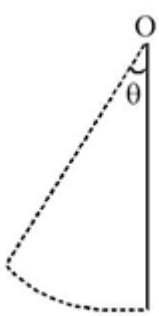
(B) $\frac{\mathrm{K}^2}{\mathrm{K}^2+\mathrm{R}^2}$

(C) $\frac{R^2}{K^2+R^2}$

(D) $\frac{\mathrm{K}^2+\mathrm{R}^2}{\mathrm{R}^2}$

Q.10

A uniform rod of length l is free to rotate in a vertical plane about a fixed horizontal axis through O. The rod begins rotating from rest from its unstable equilibrium position. When it has turned through an angle θ, its angular velocity ω is given as



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(A) $\sqrt{\frac{6\,g}{l}}\sin\theta$

(B) $\sqrt{\frac{6g}{l}}\sin\frac{\theta}{2}$

(C) $\sqrt{\frac{6g}{l}}\cos\frac{\theta}{2}$

(D) $\sqrt{\frac{6\,g}{l}}\cos\theta$

Q.11

A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is



(A) $\mathrm{g/L}$

(B) $2\ \mathrm{g/L}$

(C) $\frac{2g}{3L}$

(D) $\frac{3g}{2L}$

Q.12

A solid sphere of mass $2\ \mathrm{kg}$ rolls on a smooth horizontal surface at $10\ \mathrm{m/s}$. It then rolls up a smooth inclined plane of inclination 30° with the horizontal. Then what is the height (in meter) attained by the sphere before it stops ?

(A) 7.3

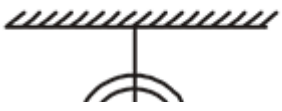
(B) 7.5

(C) 7.7

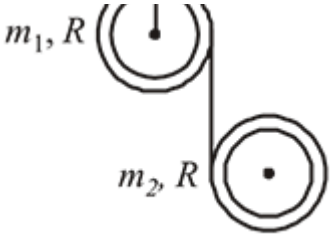
(D) 7.1

Q.13

Linear acceleration of cylinder of mass m_2 is a_2.
Then angular acceleration α_2 is (given that there is no slipping



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(A) $\frac{a_2}{R}$

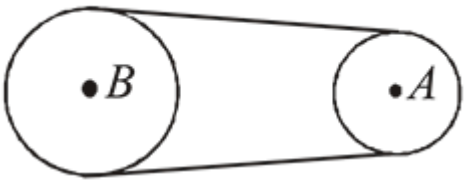
(B) $\frac{(a_2+\mathrm{g})}{R}$

(C) $\frac{2(a_2+g)}{R}$

(D) None of these

Q.14

A wheel of radius $0.1\ \mathrm{m}$(wheel A) is attached by non-stretching belt to a wheel of radius $0.2\ \mathrm{m}$ (wheel B). The belt does not slip. By the time wheel B turns through 1 revolution, wheel A will rotate through



(A) $\frac{1}{2}$

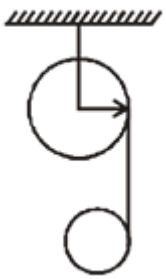
(B) 1 revolution

(C) 2 revolution

(D) 4 revolution

Q.15

Two identical discs of mass m and radius r are arranged as shown in the figure. If α is the angular acceleration of the lower disc and $\mathrm{a_{cm}}$ is acceleration of centre of mass of the lower disc, then relation between a_{cm}, α and r is



(A) $\mathrm{a_{cm}} = \alpha/\mathrm{r}$

(B) $a_{\mathrm{cm}} = 2\alpha\mathrm{r}$

(C) $\mathrm{a_{cm}} = \alpha\mathrm{r}$

(D) None of these

Q.16

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A wheel of radius $1\ \mathrm{m}$ rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is

(A) π

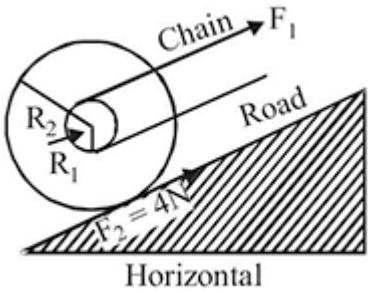
(B) 2π

(C) $\sqrt{\pi^2+9}$

(D) $\sqrt{\pi^2+4}$

Q.17

Acertain bicycle can go up a gentle incline with constant speed when the frictional force of ground pushing the rear wheel is $\mathrm{F}_2 = 4\ \mathrm{N}$. With what force F_1 (in newton) must the chain pull on Horizontal
the sprocket wheel if $R_1 = 5\ \mathrm{cm}$ and $\mathrm{R}_2 = 30\ \mathrm{cm}$?



(A) 22

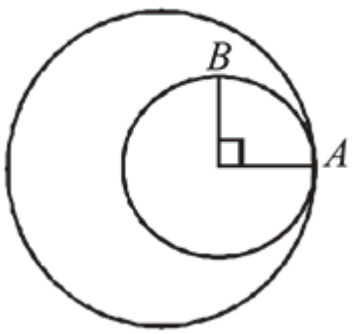
(B) 26

(C) 28

(D) 24

Q.18

A and B are moving in circular orbits with angular velocity 2ω and ω respectively. Their positions are as shown at $t = 0$. Find the time when they will meet for the first time.



(A) $\frac{\pi}{2\omega}$

(B) $\frac{3\pi}{2\omega}$

(C) $\frac{\pi}{\omega}$

(D) they will never meet

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Q.19

A fly wheel rotating about a fixed axis has a kinetic energy of 360 joule when its angular speed is 30 radian/sec. The moment of inertia of the wheel about the axis of rotation is

(A) $0.6\ \mathrm{kg/m^2}$

(B) $0.15\ \mathrm{kg\ m^2}$

(C) $0.8\ \mathrm{kg\ m^2}$

(D) $0.75\ \mathrm{kg\ m^2}$

Q.20

A bullet of $\mathrm{mass}\ 10\ \mathrm{g}$ and speed $500\ \mathrm{m/s}$ is fired into a door and gets embedded exactly at the centre of the door. The door is $1.0\ \mathrm{m}$ wide and weighs $12\ \mathrm{kg}$. It is hinged at one end and rotates about a vertical axis practically without friction. The angular speed of the door just after the bullet embeds into it will be :

(A) $6.25\mathrm{rad/sec}$

(B) $0.625\mathrm{rad/sec}$

(C) $3.35\mathrm{rad/sec}$

(D) $0.335\mathrm{rad/sec}$

Answers & Solutions

Q.1 Answer:

$\frac{v_0^2}{7g}$

Solution:

$v_p = v + R\omega = v_0 \text{ or,} at + R(\alpha t) = v_0$

$\therefore \left(\frac{2}{7}gt\right) + \frac{5}{7}gt = v_0$

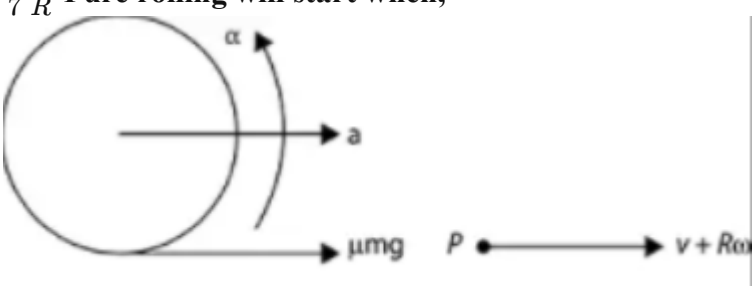
$\therefore t = \frac{v_0}{g}$

$S = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{2}{7}g\right)\left(\frac{v_0}{g}\right)^2 = \frac{v_0^2}{7g}$

$a = \frac{\mu mg}{m} = \mu g = \frac{2}{7}g$

$\alpha = \frac{\mu mg/R}{\frac{2}{5}mR^2} = \frac{5\mu g}{2R}$

$\frac{5}{7}\frac{g}{R}$ **Pure rolling will start when,**



Q.2 Answer:

$\frac{ML^2\sin^2\theta}{12}$

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Solution:

Divide the rod into a very large number of extremely small elements each of length dx. Consider one such element at a distance x from the centre O of the rod (Fig.).

Mass of element is $dm = \frac{M}{L}dx$ Perpendicular distance of the element from the axis of rotation is $r = OA = x\sin\theta$

Moment of inertia of the rod about the given axis is

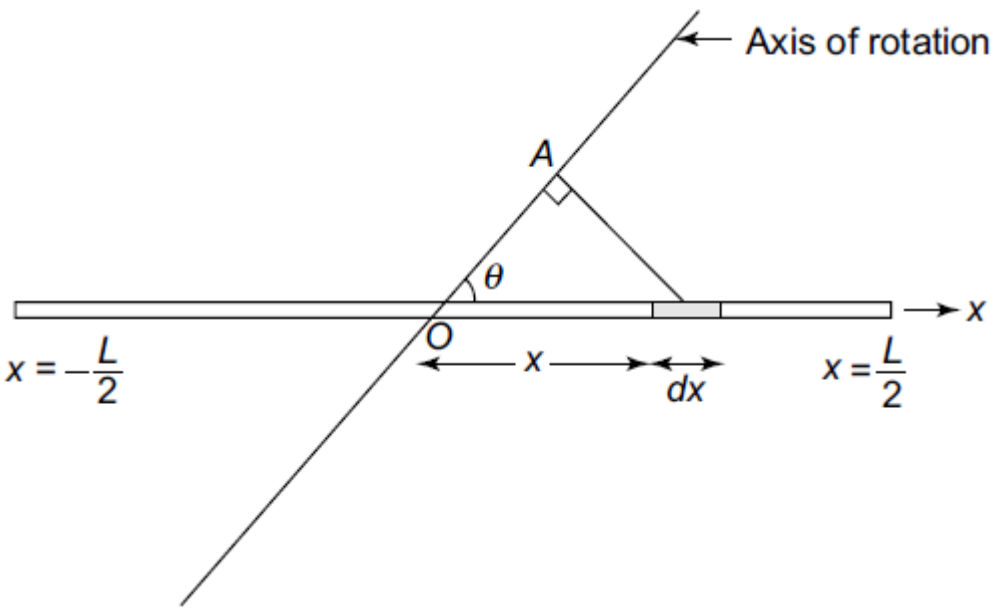
$$I = \int dmr^2 = \int \frac{M}{L}dx \times (x\sin\theta)^2$$

$$= \frac{M\sin^2\theta}{L}\int_{x=-\frac{L}{2}}^{x=+\frac{L}{2}} x^2 dx$$

$$= \frac{M\sin^2\theta}{L}\left|\frac{x^3}{3}\right|_{-L/2}^{+L/2}$$

$$= \frac{ML^2}{12}\sin^2\theta$$

$$\text{If } \theta = 90^\circ, I = \frac{ML^2}{12}$$



Q.3 Answer:

$R/\sqrt{3}$

Solution:

Let x be the distance of the axis from the centre of the sphere [Fig.]. From parallel axes theorem

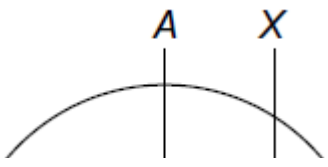
$$I_{XY} = I_{AB} + Mx^2$$

$$\Rightarrow MK^2 = \frac{2}{3}MR^2 + Mx^2$$

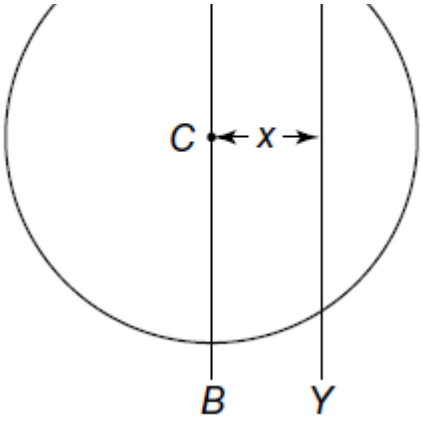
Given

$K = R$. Hence

$R^2 = \frac{2}{3}R^2 + x^2 \Rightarrow x = \frac{R}{\sqrt{3}}$



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Q.4 Answer:

$\frac{\pi}{15}\mathrm{Nm}$

Solution:

$$\tau \times \Delta \mathrm{t} = \mathrm{L}_0 \qquad \{\because \text{ since } \mathrm{L}_f = 0\}$$

or $\tau \times 30 = 2\pi$

$$\tau = \frac{\pi}{15}\ \mathrm{N-m}$$

Q.5 Answer:

$0.21\mathrm{Nm}$

Solution:

$$\tau \times \Delta \mathrm{t} = \mathrm{I}\omega$$

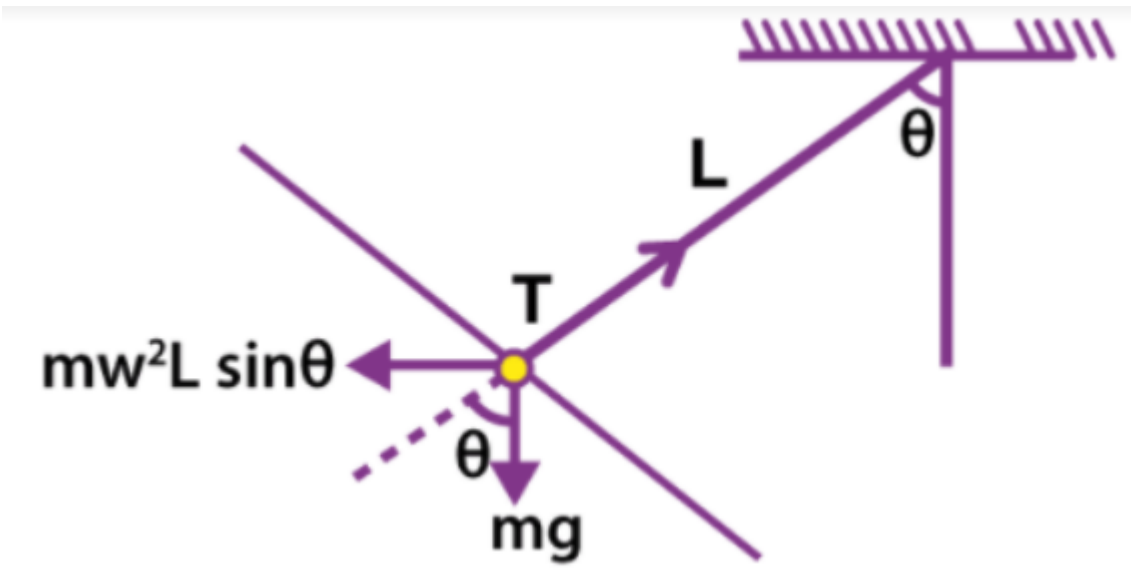
or $\tau \times 60 = 2 \times 2 \times 60\pi/60 \Rightarrow \tau = 0.21\ \mathrm{N-m}$

$\left(\because \mathrm{f} = 60\mathrm{rpm}\ \therefore \omega = 2\pi \mathrm{f} = 2\pi \times \frac{60}{60}\right)$

Q.6 Answer:

36

Solution:



$\mathrm{m}\omega^2 L \sin\theta \cos\theta = mg \sin\theta$

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$\mathrm{Cos}\,\theta = g/\omega^2 L$

$\mathrm{Sin}\,\theta = 1/\omega^2 L$

$\sin\theta = \frac{1}{\omega^2 L}\sqrt{\left(\omega^2 L\right)^2 - g^2}$

$\mathrm{T} = \mathrm{mg}\cos\theta + \mathrm{m}\omega^2 L\sin^2\theta$

$\mathrm{T} = \mathrm{mg}\left(\mathrm{g}/\omega^2 L\right) + \left(m\omega^2/\omega^2 L\right)^2\left(\left(\omega^2 L\right)^2 - g^2\right)$

$\mathrm{T} = \mathrm{mg}/\omega^2 L\left[g^2 + \left(\omega^2 L\right)^2 - g^2\right]$

$\mathrm{T} = \mathrm{m}\omega^2 L = 324(given)$

$\omega = \sqrt{\frac{324}{0.5\times 0.5}}$

$\omega = 36\mathrm{rad/s}$

Q.7 Answer:

purely rotational motion

Solution:

Couple produces purely rotational motion.

Q.8 Answer:

mass of the cube

Solution:

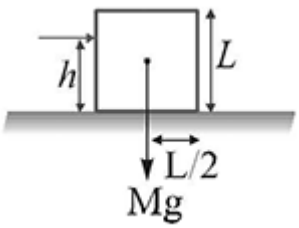
For toppling $Mg\frac{L}{2} = F_1 \times h$

For sliding $\mu Mg = F_2$

For sliding to occur first

$F_1 > F_2$

or $\frac{mgL}{2} > \mu Mg$ **or** $L > 2\mu h$



Q.9 Answer:

$\frac{\mathrm{K}^2}{\mathrm{K}^2+\mathrm{R}^2}$

Solution:

$\frac{\text{Rotational KE}}{\text{Total KE}} = \frac{\frac{1}{2}\mathrm{mv}^2\left(\frac{\mathrm{K}^2}{\mathrm{R}^2}\right)}{\frac{1}{2}\mathrm{mv}^2\left(1+\frac{\mathrm{K}^2}{\mathrm{R}^2}\right)}$

$= \frac{\mathrm{K}^2}{\mathrm{K}^2+\mathrm{R}^2}$

Q.10 Answer:

$\sqrt{\frac{6g}{l}}\cos\frac{\theta}{2}$

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Solution:

$\frac{1}{2}\mathrm{I}\omega^2 =$ **Loss of potential energy**

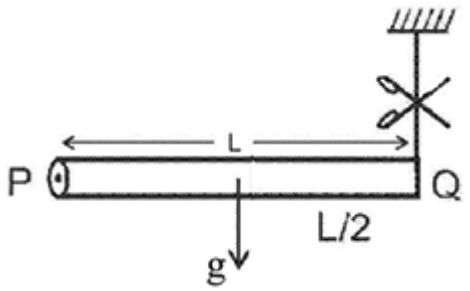
$\frac{1}{2} \times \frac{\mathrm{m}l^2}{3}\omega^2 = \frac{\mathrm{mg}l}{2}(1+\cos\theta)$

$\Rightarrow \omega^2 = \frac{3\,\mathrm{g}}{l}\left(2\cos^2\frac{\theta}{2}\right) \Rightarrow \omega = \sqrt{\frac{6\,\mathrm{g}}{l}}\cos\frac{\theta}{2}$

Q.11 Answer:

$\frac{3g}{2L}$

Solution:



angular acceleration $\alpha = \frac{3g}{2\,\mathrm{L}}$

Q.12 Answer:

7.1

Solution:

If a body rolls on a horizontal surface, it possesses both translational and rotational kinetic energies. The net kinetic energy is given by

where K **is the radius of gyration. So from law of conservation of energy,**

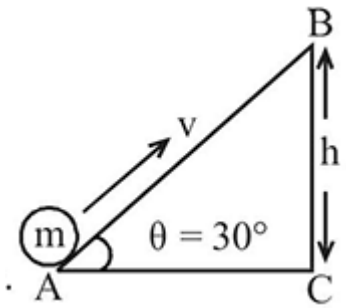
$\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right) = mgh$

where h **is the height attained by the sphere.**

i.e., $\frac{1}{2} \times 2 \times (10)^2\left(1+\frac{2}{5}\right) = 2 \times 9.8 \times \mathrm{h}.$

i.e., $\frac{1}{2} \times 100 \times \left(\frac{7}{5}\right) = 9.8\,\mathrm{h}$

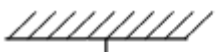
or $\mathrm{h} = \frac{700}{98} = 7.1\,\mathrm{m}$



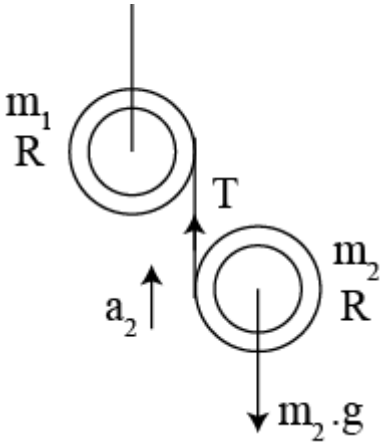
Q.13 Answer:

$\frac{a_2}{R}$

Solution:



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$a_2 \rightarrow$ **linear acceleration of mass** m_2 $\alpha_2 \rightarrow$ **angular acceleration of** m_2 $\alpha_1 \rightarrow$ **angular acceleration of** m_1 **Since no external force, net force will be zero.** $m_2 \cdot g - T - m_2 a_2 = 0.$
We know $T \cdot R = I\alpha [T \rightarrow$ **torque;** $R \rightarrow$ **radius** $] I \rightarrow$ **moment of inertia** $= mR^2/2$ **for cylinder** $T \cdot R = I_2 \alpha_2 \quad [T = m_2 g - m_2 \cdot a_2] \Rightarrow \left(m_2 \cdot g - m_2 \cdot a_2\right) R = I_2 \left(\frac{a_2}{R}\right) \quad [\alpha = a/R] \Rightarrow m_2 \left(g - a_2\right) = \frac{m_2 R^2 \left(\frac{a_2}{R^2}\right)}{2} = a_2 + \frac{a_2}{2} = \frac{3a_2}{2} \Rightarrow g - a_2 = \frac{a_2}{2} \Rightarrow g = \alpha_2$ **Since there is no slipping** $\Rightarrow \alpha_1 = \alpha_2$ $\alpha_2 = \frac{a_2}{R} = \frac{2g}{3R} \Rightarrow \alpha_1 = \frac{2g}{3R} = \frac{a_2}{R}$
Option (A) **is the correct answer.**

Q.14 Answer:

2 revolution

Solution:

Let speed of belt be v **Angular speeds of wheels** $\omega_B = \frac{v}{2\pi R_B}, \omega_A = \frac{v}{2\pi R_A}; \frac{\omega_A}{\omega_B} = \frac{R_B}{R_A} = 2$

Q.15 Answer:

$a_{\text{cm}} = 2\alpha \text{r}$

Solution:

Tr $= \frac{mr^2}{2}\alpha_1 \quad \ldots\ldots\ldots$
(2) $\text{Tr} = \frac{\text{mr}^2}{2}\alpha \quad \ldots\ldots$
(3) $\alpha_1 = \alpha$
$\ldots\ldots$
Acceleration of point $\text{b} =$ **acceleration of point a** $r\alpha_1 = a_{\text{cm}} - r\alpha$
$\ldots\ldots\ldots(4)$
Hence, $2\text{r}\alpha = \text{a}_{\text{cm}}$

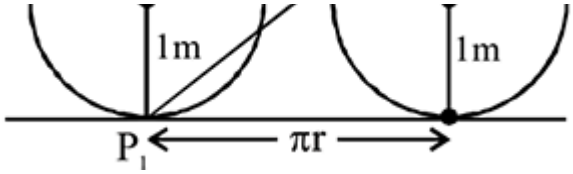
Q.16 Answer:

$\sqrt{\pi^2 + 4}$

Solution:



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Linear distance moved by wheel in half revolution $= \pi r$. Point P_1 after half revolution reaches at P_2 vertically 2 m above the ground.

$\therefore$ Displacement $P_1P_2 = \sqrt{\pi^2 r^2 + 2^2} = \sqrt{\pi^2 + 4} \quad [\because r = 1m]$

Q.17 Answer:

24

Solution:

(24) For no angular acceleration $\tau_{\text{net}} = 0$

$\Rightarrow \mathrm{F}_1 \times 5 = \mathrm{F}_2 \times 30$ (**given** $\mathrm{F}_2 = 4$ N) $\Rightarrow \mathrm{F}_1 = 24$ N

Q.18 Answer:

they will never meet

Solution:

Case 1: When they rotate in same sense $2m\pi = 2\omega t$

$\frac{3\pi}{2} + 2n\pi = \omega t; 2m\pi = 2\left(\frac{3\pi}{2} + 2n\pi\right)$

$2m = 3 + 4n; \quad m = \frac{3}{2} + 2n \Rightarrow m - 2n = \frac{3}{2}$

Not possible for m **and** n **being integer. Case** 2 **: When they rotate in opposite sense** $2m\pi = 2\omega t$

$\frac{\pi}{2} + 2n\pi = \omega t; 2m\pi = 2\left(\frac{\pi}{2} + 2n\pi\right)$

$2m\pi = \pi + 4n\pi; 2m - 4n = 1$

Not possible for m **and** n **integer.**

Q.19 Answer:

0.8 kg m^2

Solution:

$E_r = \frac{1}{2} I\omega^2 \Rightarrow$

$I = \frac{2E_r}{\omega^2} = \frac{2\times 360}{30\times 30} = 0.8$ kg m^2

Q.20 Answer:

$0.625 \mathrm{rad/sec}$

Solution:

Let x $=$ **distance between hinge and centre of the door** $= \mathrm{L}/2 = \frac{1}{2} = 0.5$ m

As per conservation of angular momentum

$\mathrm{mvx} = \mathrm{I}\omega$

$\Rightarrow \omega = \frac{mvx}{I} = \frac{0.01\times 500\times 0.5}{\frac{1}{3}\times 12\times (1)^2} = 0.625 \mathrm{rad/s}$

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